

DK LEARNING

GRADE 6

Networks & Problem Solving

Level: **Intermediate** • Purpose: Learn before you practise

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In this guide

1. What is a Network?
2. LAN, WAN and the Internet
3. Network Devices
4. Troubleshooting Step by Step
5. Staying Secure
6. Practice questions, answers and glossary



How to use this guide

Read each chapter slowly. Look at the example, then try the “Your turn” task **without copying**. Write your own notes on the lines. When you finish, answer the practice questions and check your work with the answer page. After that, open the Grade 6 activity on the website.

Learning goals

- Explain what a network is and name network devices
- Describe LAN, WAN and the Internet
- Follow step-by-step troubleshooting
- Use good security practices

Study plan

Step	What to do
1	Read the chapter and highlight key words
2	Study the worked example
3	Do the “Your turn” task on paper or on a computer
4	Write a short summary in your own words
5	Answer the practice questions, then check

Tip: When something fails, change only one thing at a time. That way you know what fixed it.

1 What is a Network?

A network connects devices so they can share information and resources. A local network can connect computers, printers and other devices in a school or home.

Networks let people share files, use one printer, play together and access the Internet.

Key points

- Network = connected devices
- Used to share files and printers
- Wired or wireless
- Many devices, one connection

Worked example

School lab:

20 computers + 1 printer + 1 router = a local network

Your turn

Draw a simple home network with a router, phone, laptop and printer.

Check yourself

Explain each term in your own words:

Network

Resource

My notes

2 LAN, WAN and the Internet

A LAN (local area network) covers a small area such as a room or school. A WAN (wide area network) covers a large area such as a country. The Internet is a worldwide network of networks.

A router connects your LAN to other networks, including the Internet.

Key points

- LAN = small area
- WAN = large area
- Internet = network of networks
- Router connects networks

Worked example

Your home Wi-Fi = LAN

A bank linking branches across Sri Lanka = WAN

Your turn

Classify: school lab, a global company network, your home network.

Check yourself

Explain each term in your own words:

LAN

WAN

My notes

3 Network Devices

A router sends data between networks. A switch connects wired devices in one LAN. A wireless access point lets devices connect by Wi-Fi. A network cable (Ethernet) carries data in a wired network.

Each device on a network has an address so data reaches the right place.

Key points

- Router: between networks
- Switch: connects wired devices
- Access point: Wi-Fi
- Ethernet cable: wired link

Worked example

Laptop -> Wi-Fi -> Router -> Internet -> Website

Your turn

Write the job of a router, a switch and an access point in one sentence each.

Check yourself

Explain each term in your own words:

Router

Switch

My notes

4 Troubleshooting Step by Step

Troubleshooting means finding the cause of a problem step by step. Start with simple checks such as power, cables, Wi-Fi and error messages. Change one thing at a time and test after each change.

Write down what you tried so you can undo a mistake.

Key points

- Check the simplest things first
- Read error messages
- One change at a time
- Restart can fix many problems

Worked example

Problem: Web page will not load.

1. Is Wi-Fi connected?
2. Do other sites work?
3. Restart the router.
4. Try again.

Your turn

Write a troubleshooting checklist for a printer that will not print.

Check yourself

Explain each term in your own words:

Troubleshooting

Error message

My notes

5 Staying Secure

Security practices include strong passwords, updates, safe links and secure connections. Multi-factor authentication adds a second check, such as a code sent to a phone.

Public Wi-Fi may not be safe. Avoid entering passwords on unknown networks.

Key points

- Use strong, unique passwords
- Install updates
- Do not click strange links
- Use multi-factor authentication

Worked example

Login steps with MFA:

1. Enter password.
2. Enter the code sent to your phone.

Your turn

Explain why software updates can improve security.

Check yourself

Explain each term in your own words:

MFA

Update

My notes



Practice Questions

Part A - Choose the best answer

1. A network is...
 - A) connected devices sharing information
 - B) a single computer
 - C) a printer cable only
2. Which covers a small area?
 - A) LAN
 - B) WAN
 - C) Internet
3. Which device connects networks?
 - A) Router
 - B) Mouse
 - C) Speaker
4. First step in troubleshooting?
 - A) Check simple things
 - B) Throw the device
 - C) Buy new one
5. MFA means...
 - A) a second check at login
 - B) a type of cable
 - C) a font
6. Why install updates?
 - A) They fix weaknesses
 - B) They make the screen red
 - C) They delete files

Part B - Short answers

1. What is a router?

2. Give one example of a LAN.

3. Why change only one thing at a time?

4. Name one secure habit.



Answer Page

Part A

1-A 2-A 3-A 4-A 5-A 6-A

Part B (sample answers)

1. A device that forwards data between networks.
2. A school computer lab or home Wi-Fi.
3. To know which change fixed the problem.
4. Strong passwords, updates, MFA or safe links.

Mini glossary

Network	A group of connected devices
Resource	Something shared, like a printer
LAN	Local area network
WAN	Wide area network
Router	Device that forwards data between networks
Switch	Device that links wired devices in a LAN
Troubleshooting	Finding and fixing a problem
Error message	A message explaining what went wrong
MFA	Using two or more checks to log in
Update	A new version that fixes problems and weaknesses



Activity Preparation

Solve network troubleshooting cases and identify secure responses. Then open the Grade 6 activity.

Think before you start

- Explain one important idea from this grade in your own words.
- Give one real-life example where this topic is useful.
- What mistake could a beginner make, and how can it be prevented?
- Write the steps you will follow before starting the Grade 6 activity.

Ready checklist

- | | |
|---|--|
| ☐ I read every chapter | ☐ I tried the “Your turn” tasks |
| ☐ I understand the key terms | ☐ I checked my practice answers |
| ☐ I wrote my own notes | ☐ I am ready for the activity |

⚡ **START GRADE 6 ACTIVITY**

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